

B. Niko's Tactical Cards

time limit per test: 1.5 seconds
 memory limit per test: 256 megabytes

Niko is playing a game. Her score is denoted by an integer k which is 0 initially.

The game has n turns. On the i -th turn, Niko is given a red card with an integer a_i on it, as well as a blue card with an integer b_i on it. She must choose exactly **one** of the cards and update her score according to her choice:

- If she chooses the red card, her score becomes $k - a_i$, where k is her score before the turn.
- If she chooses the blue card, her score becomes $b_i - k$, where k is her score before the turn.

After this, the game proceeds to the next turn, or ends if it is the n -th turn.

Your task is to find the maximum possible score Niko can obtain at the end of the game.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \leq t \leq 10^3$). The description of the test cases follows.

The first line of each test case contains a single integer n ($1 \leq n \leq 10^5$) — the number of turns.

The second line of each test case contains n integers $a_1, a_2, \dots, a_n$ ($-10^9 \leq a_i \leq 10^9$).

The third line of each test case contains n integers $b_1, b_2, \dots, b_n$ ($-10^9 \leq b_i \leq 10^9$).

It is guaranteed that the sum of n over all test cases does not exceed 10^5 .

Output

For each test case, output a single integer — the maximum possible score Niko can obtain at the end of the game.

Example

input	Copy
3	
3	
4 -8 -1	
-3 -7 0	
5	
-3 1 0 7 1	
-5 3 -1 4 -5	
5	
-7 7 5 4 9	
-9 -3 3 2 2	
output	Copy
6	
12	
27	

Note

In the first test case, one optimal strategy is as follows:

Codeforces Round 1068 (Div. 2)

Contest is running

01:37:33

Contestant



→ Submit?

Language: GNU G++17 7.3.0

Choose file: No file chosen

Be careful: there is 50 points penalty for submission which fails the pretests or resubmission (except failure on the first test, denial of judgement or similar verdicts).
 "Passed pretests" submission verdict doesn't guarantee that the solution is absolutely correct and it will pass system tests.

→ Score table

	Score
Problem A	456
Problem B	912
Problem C	1140
Problem D	1596
Problem E	2052
Problem F	2736
Successful hack	100
Unsuccessful hack	-50
Unsuccessful submission	-50
Resubmission	-50

* If you solve problem on 00:22 from the first attempt

Turn	0	1	2	3
Card Chosen	—	Blue	Red	Red
Score	0	$-3 - 0 = -3$	$-3 - (-8) = 5$	$5 - (-1) = 6$

In the second test case, one optimal strategy is as follows:

Turn	0	1	2	3	4	5
Card Chosen	—	Blue	Blue	Blue	Blue	Red
Score	0	-5	8	-9	13	12

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